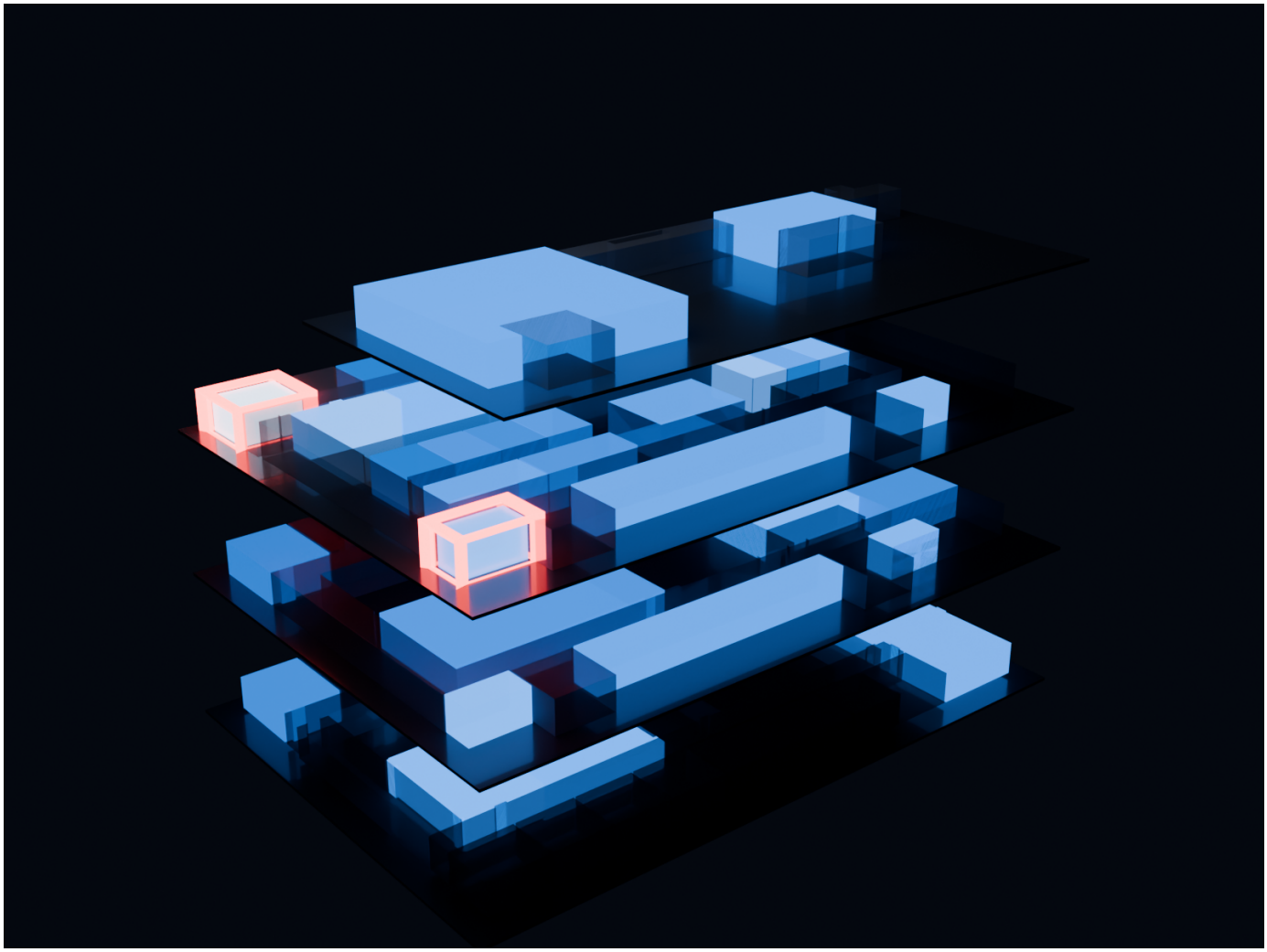
SIMULATED SENSOR DATA · physics-based feed with planted faults · Building 308 renovation design model

Indeklima Twin: does every room keep its design promise?

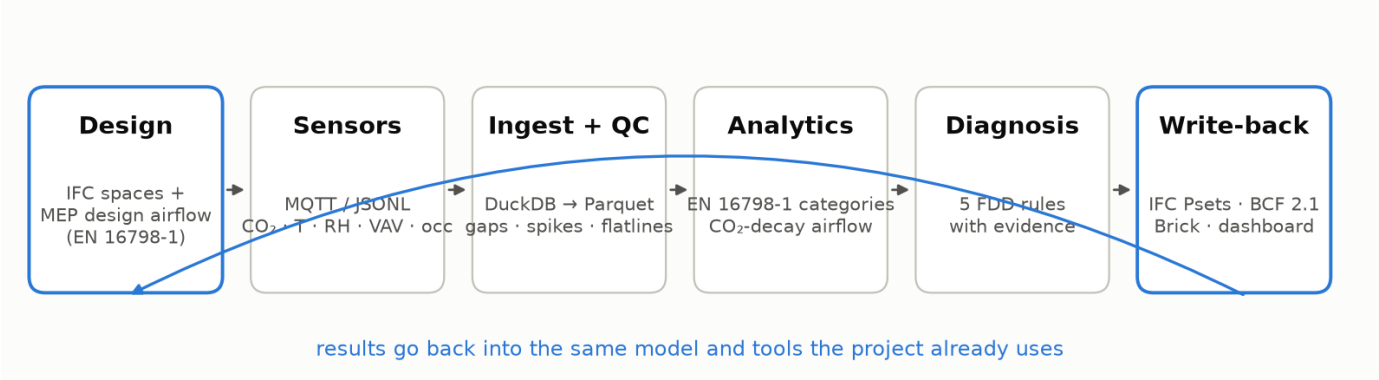
Abdallah Alhussein · MSc Architectural Engineering · indoor climate, building physics, data pipelines



The renovation design already states what each room should do: how many people, how much outdoor air, which EN 16798-1 category. Indeklima Twin reads that promise from the project's IFC model and MEP design, runs the sensor feed through quality control, comfort analysis, CO_2-decay airflow inference and fault diagnosis, and writes the verdict back into the IFC, BCF and Brick files the project already works with.

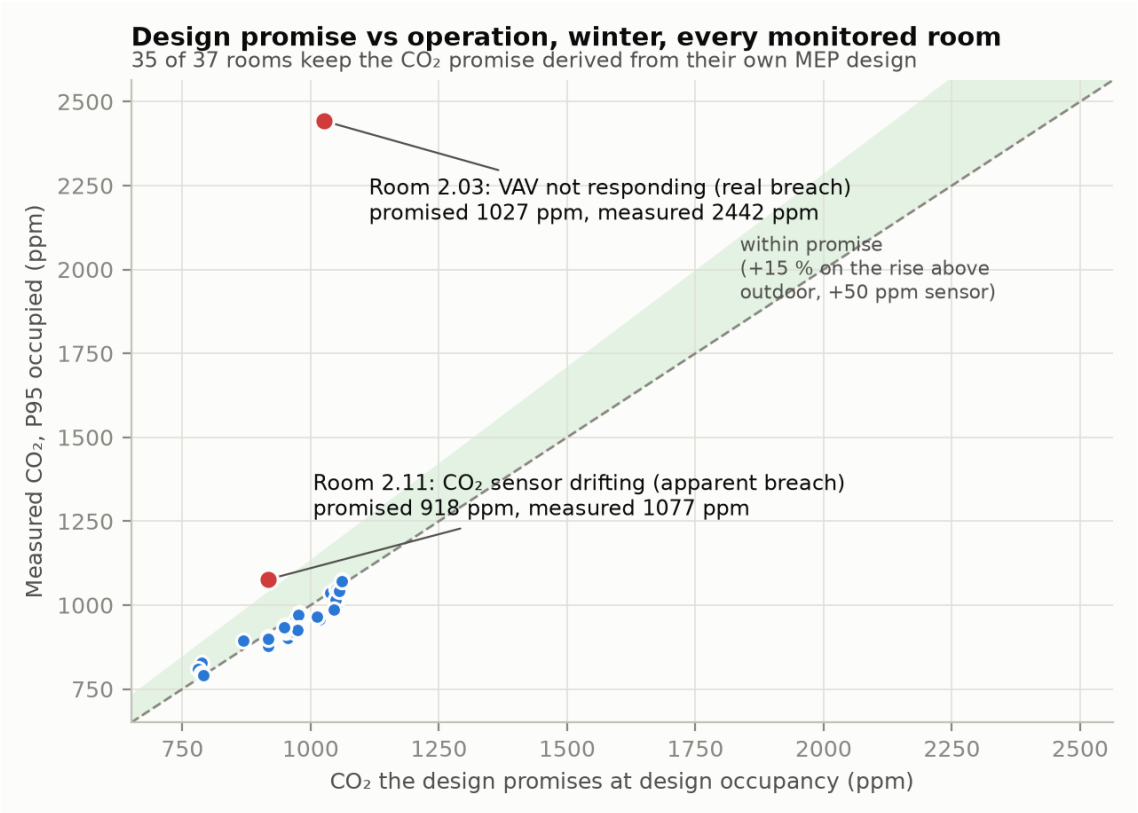
Rooms twinned	Keep the CO2 promise (winter)	Keep both promises (both seasons)	Faults diagnosed	False alarms	Airflow error
37	35/37	16/37	5 + 2 data	0	9.9 %

1. How it works

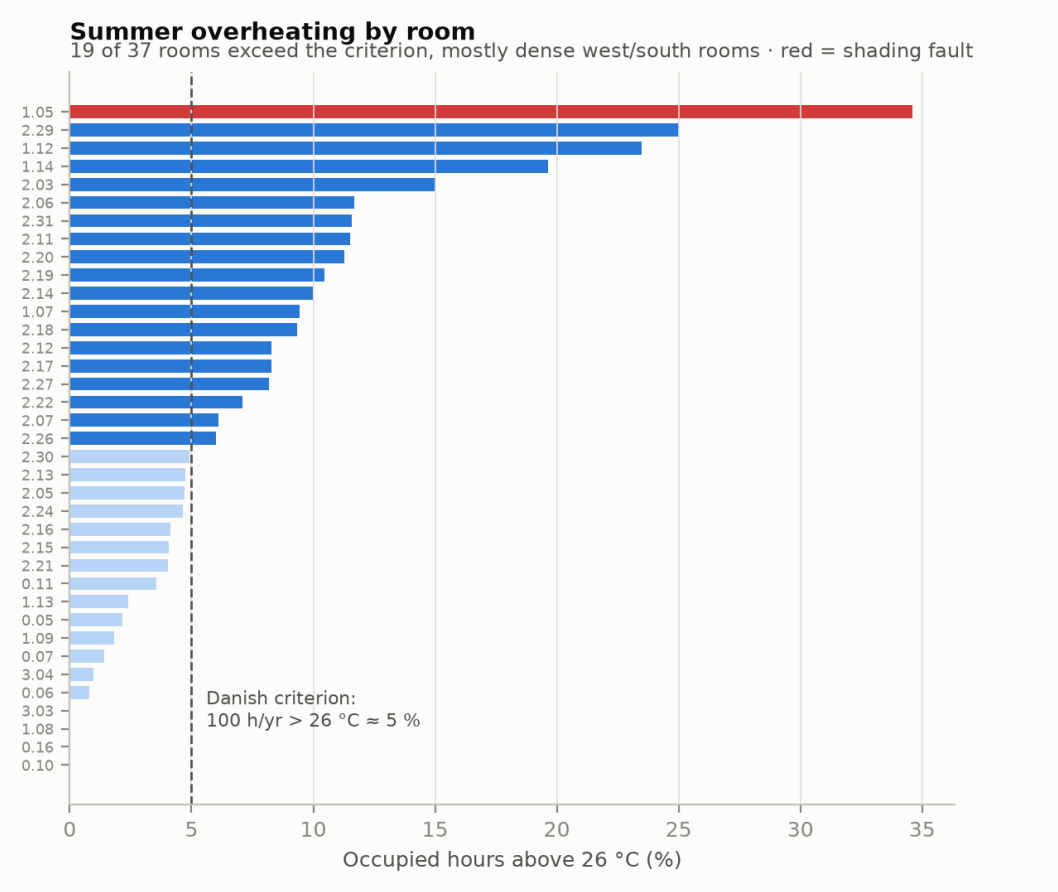


Design. 37 occupied rooms (offices, meeting rooms, multipurpose rooms, atrium, auditorium) are taken from the project's IFC4 model (IfcSpace geometry, areas, volumes). Design airflow follows the project's MEP method: 0.7 L/s·m² + 7 L/s per person (EN 16798-1 cat. II). The promised CO_2 level is the steady state at design occupancy, $C = C_{out} + N \cdot G/q$. **Sensors.** 896,724 MQTT-style messages (CO_2, temperature, RH, VAV airflow, occupancy, weather) for two winter and two summer weeks. **QC.** Range checks, a Hampel spike filter, gap and flatline detection (403 spikes removed, completeness 99.3 %). **Analytics.** EN 16798-1 categories over occupied hours; outdoor-air change rate from CO_2 decay when a room empties. **Diagnosis.** Five rule families with evidence. **Write-back.** 37 IfcSpaces enriched (standard IFC4 property sets for the design intent plus an operations set), 7 BCF 2.1 topics, a Brick model of 1812 triples.

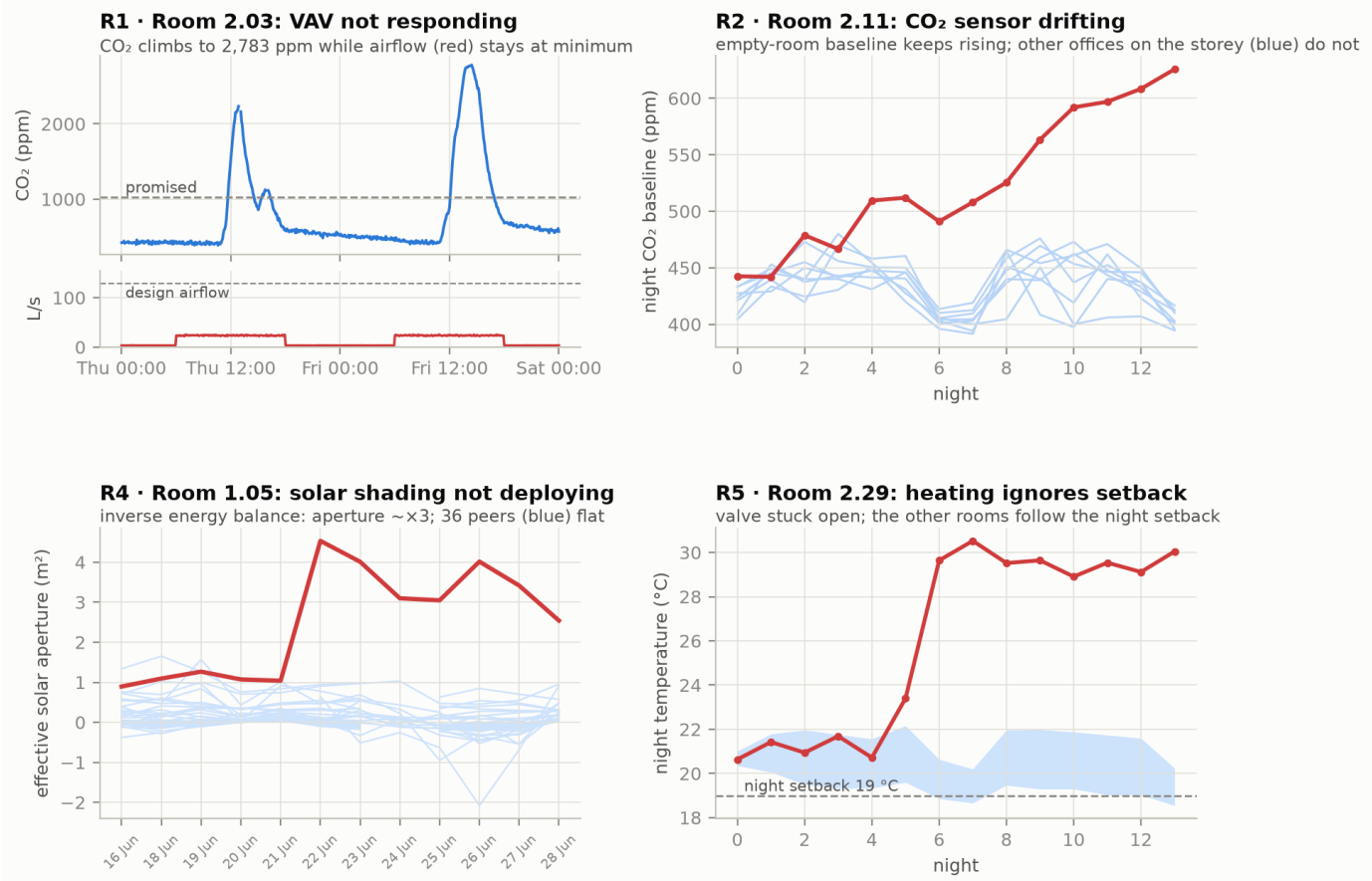
2. Promise vs operation



Every room is judged against what its own design promised, not a generic threshold. Two rooms break the winter CO_2 promise, and the diagnosis separates them: room 2.03 really is under-ventilated (VAV not responding), while room 2.11 only *appears* to be: its CO_2 sensor is drifting. The temperature promise is stricter: in summer many small west and south rooms exceed the Danish practice of at most 100 h/year above 26 °C, the same design-stage risk the IDA ICE study flagged.


Summer overheating by room
19 of 37 rooms exceed the criterion, mostly dense west/south rooms · red = shading fault
1.05
2.29
1.12
1.14
2.03
2.06
2.31
2.11
2.20
2.19
2.14
1.07
2.18
2.12
2.17
2.27
2.22
2.07
2.26
2.30
2.13
2.05
2.24
2.16
2.15
2.21
0.11
1.13
0.05
1.09
0.07
3.04
0.06
3.03
1.08
0.16
0.10
Danish criterion:
100 h/yr > 26 °C ≈ 5 %
0
5
10
15
20
25
30
35
Occupied hours above 26 °C (%)

3. Findings, each with its evidence

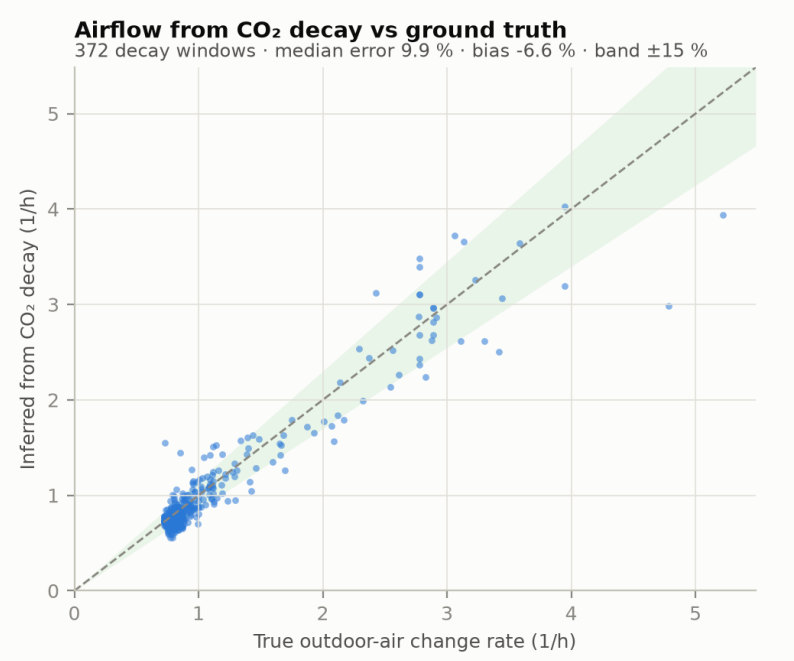


#	Finding	Room	Evidence	Diagnosis
R1	Airflow not responding to demand	2.03	CO2 up to 2783 ppm while airflow stayed at 24 L/s (min 24, design 129 L/s) on 7 days	VAV damper / actuator not responding to CO2 demand
R2	CO2 sensor drifting	2.11	night baseline rose from 442 to 625 ppm (14.3 ppm/day, R2 0.94) in an empty room	CO2 sensor drift: recalibrate / replace (VAV is over-ventilating on it)
R3	AHU zone running out of hours	storey 3	100 % of storey-3 rooms ventilated at 284 L/s median with nobody in, 7 nights; ~96 kWh fan energy wasted	AHU zone schedule overridden (manual / BMS time program)
R4	Solar shading not deploying	1.05	effective solar aperture (inverse energy balance) rose from 1.1 to 3.4 m2 (x3.2) from 2026-06-22; peer rooms unchanged (median x0.53)	external solar shading not deploying (motor / control / wind lock)
R5	Heating ignores night setback	2.29	night temperature 29.6 C on 8 nights (setback 19 C; other rooms 20.8 C the same nights)	radiator valve / thermostat stuck open
QC	Gateway offline	storey 2	gateway outage for 6.0 h on 22 rooms	data issue: not reported as a building fault
QC	Sensor frozen	3.04	flatline for 70.0 h	data issue: not reported as a building fault

Data problems (a frozen sensor, an offline gateway) are reported by QC and are never presented as building faults, so nobody is sent to inspect a healthy VAV.

4. Validation

The sensor feed is simulated, so the truth is known. The detectors never see it. **Faults:** five building / sensor faults and two data faults are planted per run. On 11 development runs, 11/11 found all seven with no false alarm. On 5 **held-out** seeds, never used while designing the rules, 5/5 did. A fault-free building raised 0 findings and 0 QC alarms. **Airflow:** CO_2-decay estimates vs the true outdoor-air rate: median error 9.9 %, P90 22 %, bias -6.6 % (the CO_2 sensor's lag and gain error, both stated). **Physics:** the CO_2 balance is integrated exactly and tested against the closed-form solution (< 1 ppm) and the steady state (< 0.5 %).



Seed	Held out	Building / sensor faults	Data faults	False alarms	Airflow error
1		5/5	2/2	0	11.0 %
2		5/5	2/2	0	9.6 %
3		5/5	2/2	0	9.8 %
4		5/5	2/2	0	9.6 %
5		5/5	2/2	0	9.0 %
6		5/5	2/2	0	10.5 %
7		5/5	2/2	0	8.1 %
8		5/5	2/2	0	9.3 %
9		5/5	2/2	0	10.4 %
10		5/5	2/2	0	9.0 %
11	yes	5/5	2/2	0	10.0 %
12	yes	5/5	2/2	0	9.9 %
13	yes	5/5	2/2	0	11.2 %
14	yes	5/5	2/2	0	10.0 %
15	yes	5/5	2/2	0	10.0 %
308		5/5	2/2	0	9.9 %

5. Limits

Simulated, not measured data: the pipeline accepts a real MQTT / CSV feed unchanged, and a pilot building is the next step. The synthetic feed uses a single-zone (1R1C) thermal model per room; the detectors do not depend on it. Detector thresholds were developed on this building and are stated in the code. Five rule families, not a full rule library. Read-only: it diagnoses and does not control equipment.